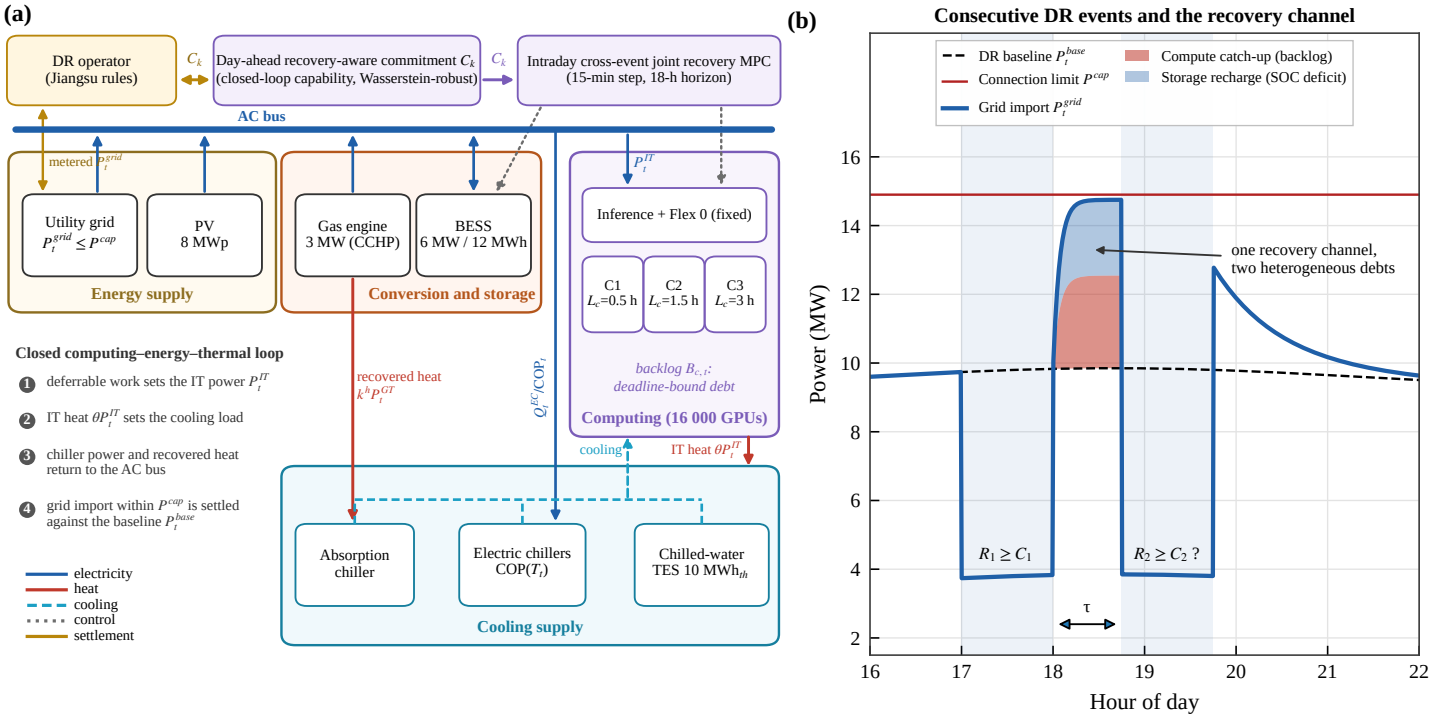

(a)
DR operator
(Jiangsu rules)
C_k
Day-ahead recovery-aware commitment C_k
(closed-loop capability, Wasserstein-robust)
C_k
Intraday cross-event joint recovery MPC
(15-min step, 18-h horizon)
AC bus
metered P_t^{grid}
Utility grid
$P_t^{grid} \le P^{cap}$
PV
8 MWp
Energy supply
Gas engine
3 MW (CCHP)
BESS
6 MW / 12 MWh
Conversion and storage
P_t^{IT}
Inference + Flex 0 (fixed)
C1
L_c=0.5 h
C2
L_c=1.5 h
C3
L_c=3 h
backlog $B_{c,t}$:
deadline-bound debt
Computing (16 000 GPUs)
recovered heat
$k^h P_t^{GT}$
Q_t^{EC}/COP_t
cooling
IT heat θP_t^{IT}
Absorption
chiller
Electric chillers
COP(T_t)
Chilled-water
TES 10 MWh$_{th}$
Cooling supply
Closed computing–energy–thermal loop
1 deferrable work sets the IT power P_t^{IT}
2 IT heat θP_t^{IT} sets the cooling load
3 chiller power and recovered heat return to the AC bus
4 grid import within P^{cap} is settled against the baseline P_t^{base}
electricity
heat
cooling
control
settlement
(b)
Consecutive DR events and the recovery channel
DR baseline P_t^{base}
Connection limit P^{cap}
Grid import P_t^{grid}
Compute catch-up (backlog)
Storage recharge (SOC deficit)
one recovery channel,
two heterogeneous debts
$R_1 \ge C_1$
$R_2 \ge C_2$?
τ
Power (MW)
Hour of day
16
14
12
10
8
6
4
2
16
17
18
19
20
21
22